

SISTEMA FV CONECTADO A RED

# PVGISAMPLIADOMULTI

Mi rrrr Real

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*18 de June de 2026*

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## Resumen

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum defuturum, quas natura non depravata desiderat. Et quem ad me accedis, saluto: 'chaere,' inquam, 'Tite!' lictores, turma omnis chorusque: 'chaere, Tite!' hinc hostis mi Albucius, hinc inimicus. Sed iure Mucius. Ego autem mirari satis non queo unde hoc sit tam insolens domesticarum rerum fastidium. Non est omnino hic docendi locus; sed ita prorsus existimo, neque eum Torquatum, qui hoc primus cognomen invenerit, aut torquem illum hosti detraxisse, ut aliquam ex eo est consecutus? – Laudem et caritatem, quae sunt vitae sine metu degendae praesidia firmissima. – Filium morte multavit. – Si sine causa, nollem me ab eo delectari, quod ista Platonis, Aristoteli, Theophrasti orationis ornamenta neglexerit. Nam illud quidem physici, credere aliquid esse minimum, quod profecto numquam putavisset, si a Polyaeo, familiari suo, geometrica discere maluisset quam illum etiam ipsum dedocere. Sol Democrito magnus videtur, quippe homini erudito in geometriaque perfecto, huic pedalis fortasse; tantum enim esse omnino in nostris poetis aut inertissimae segnitiae est aut fastidii delicatissimi. Mihi quidem.*

# pvgisampliadomulti

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## Glosario de símbolos / nomenclatura

| Variable    | Descripción                                                              | Unidad                 |
|-------------|--------------------------------------------------------------------------|------------------------|
| E_d         | Producción de energía diaria promedio del sistema                        | kWh/d                  |
| E_m         | Producción de energía mensual promedio del sistema                       | kWh/mo                 |
| E_y         | Producción de energía anual promedio del sistema                         | kWh/y                  |
| H_d         | Irradiación global diaria promedio por m <sup>2</sup> sobre los módulos  | kWh/m <sup>2</sup> /d  |
| H_m         | Irradiación global mensual promedio por m <sup>2</sup> sobre los módulos | kWh/m <sup>2</sup> /mo |
| H_y         | Irradiación global anual promedio por m <sup>2</sup> sobre los módulos   | kWh/m <sup>2</sup> /y  |
| SD_m        | Desviación estándar de la producción mensual (variación interanual)      | kWh                    |
| SD_y        | Desviación estándar de la producción anual (variación interanual)        | kWh                    |
| l_aoi       | Pérdida por ángulo de incidencia                                         | %                      |
| l_spec      | Pérdida espectral                                                        | %                      |
| l_tg        | Pérdida por temperatura e irradiancia                                    | %                      |
| l_total     | Pérdida Total del sistema                                                | %                      |
| LCOE_pv     | Costo nivelado de la electricidad fotovoltaica                           | moneda/kWh             |
| System Cost | Costo Total de inversión del sistema                                     | moneda                 |
| Interest    | Tasa de interés anual aplicada al capital                                | %/y                    |
| Lifetime    | Vida útil estimada del sistema fotovoltaico                              | años                   |

Tabla 1: Glosario de símbolos / nomenclatura

1. Grupo FV G2.



Figura 1: (G2) Paneles FV.

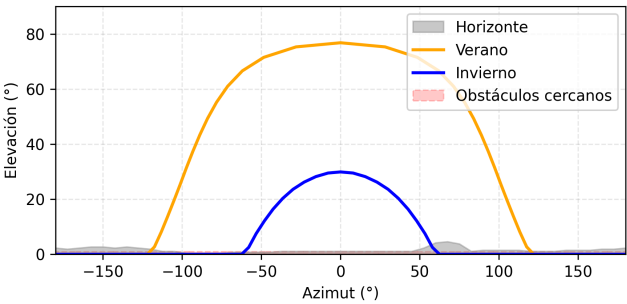


Figura 2: (G2) Perfil del horizonte y sombras.

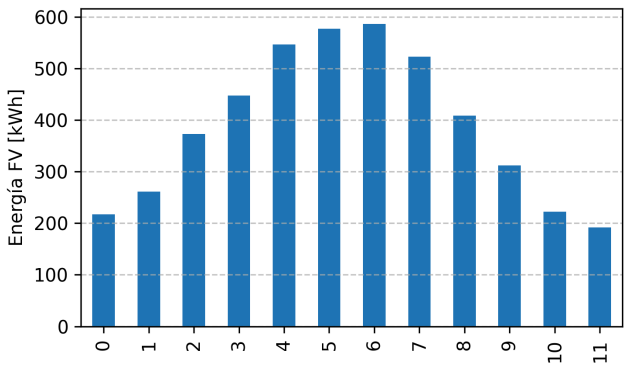


Figura 3: (G2) Producción de energía mensual.

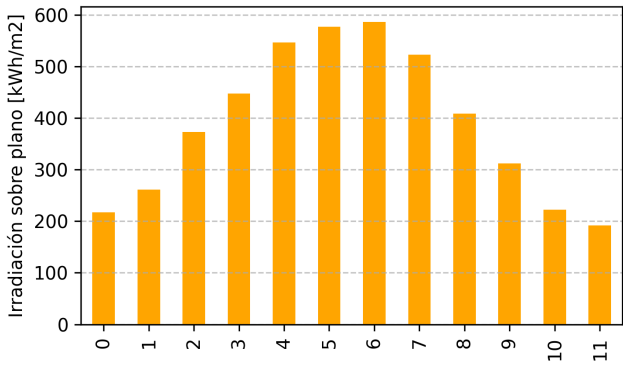


Figura 4: (G2) Irradiación mensual sobre plano fijo.

| Parámetro        | Valor          |
|------------------|----------------|
| Latitud/Longitud | 36.667, -4.461 |
| Horizonte        | DEM-calculated |
| Base de datos    | PVGIS-SARAH3   |
| Tecnología FV    | c-Si           |
| FV instalado     | 3.24 kWp       |
| Pérdidas sistema | 14.0%          |
| Costo sistema    | 3240.0 €       |
| Interés anual    | 5.0%           |
| Vida útil        | 20 años        |

Tabla 2: (G2) Datos proporcionados

| Parámetro                   | Valor                      |
|-----------------------------|----------------------------|
| Ángulo de inclinación       | 0°                         |
| Ángulo de azimut            |                            |
| Producción diaria (E_d)     | 12.78 kWh                  |
| Producción mensual (E_m)    | 388.83 kWh                 |
| Producción anual (E_y)      | 4665.96 kWh                |
| Irradiación diaria (H_d)    | 5.15 kWh/m <sup>2</sup>    |
| Irradiación mensual (H_m)   | 156.72 kWh/m <sup>2</sup>  |
| Irradiación anual (H_y)     | 1880.65 kWh/m <sup>2</sup> |
| Desv. estándar mensual      | 7.61 kWh                   |
| Desv. estándar anual        | 91.26 kWh                  |
| Pérdida ángulo incidencia   | -3.51%                     |
| Pérdida espectral           | ?(0)                       |
| Pérdida temp. e irradiancia | -7.72%                     |
| Pérdidas totales            | -23.42%                    |
| LCOE                        | 0.07 €/kWh                 |

Tabla 3: (G2) Resultados de la simulación

| month | E_d   | E_m    | SD_m  |
|-------|-------|--------|-------|
| 1     | 7.01  | 217.24 | 19.5  |
| 2     | 9.34  | 261.57 | 24.72 |
| 3     | 12.02 | 372.54 | 42.92 |
| 4     | 14.91 | 447.36 | 33.96 |
| 5     | 17.62 | 546.33 | 36.88 |
| 6     | 19.24 | 577.18 | 18.89 |
| 7     | 18.92 | 586.51 | 12.13 |
| 8     | 16.85 | 522.46 | 15.38 |
| 9     | 13.62 | 408.54 | 17.86 |
| 10    | 10.06 | 311.99 | 22.71 |
| 11    | 7.42  | 222.47 | 19.18 |
| 12    | 6.19  | 191.77 | 15.48 |

Tabla 4: (G2) Energía FV mensual

2. Grupo FV G3.



Figura 5: (G3) Paneles FV.

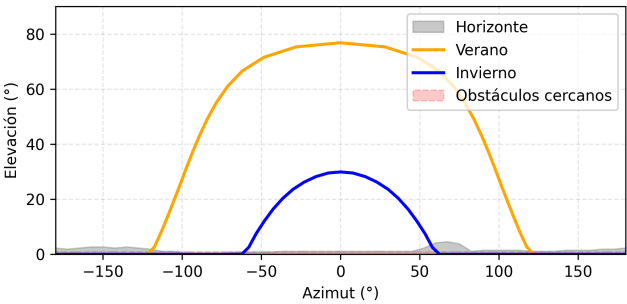


Figura 6: (G3) Perfil del horizonte y sombras.

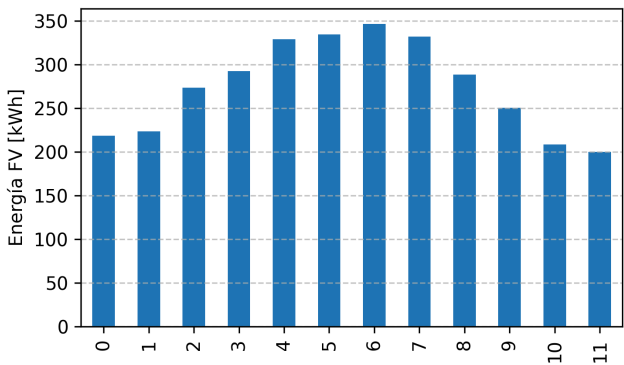


Figura 7: (G3) Producción de energía mensual.

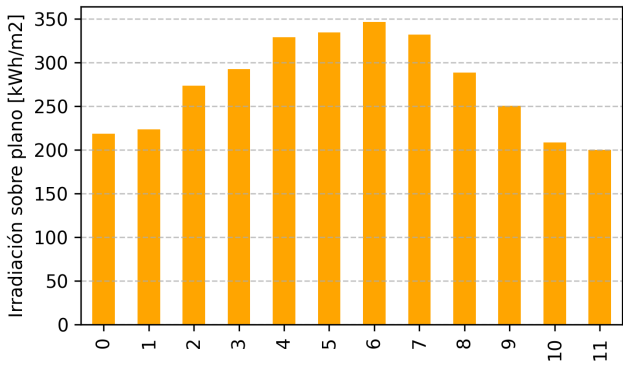


Figura 8: (G3) Irradiación mensual sobre plano fijo.

| Parámetro        | Valor          |
|------------------|----------------|
| Latitud/Longitud | 36.667, -4.461 |
| Horizonte        | DEM-calculated |
| Base de datos    | PVGIS-SARAH3   |
| Tecnología FV    | c-Si           |
| FV instalado     | 2.0 kWp        |
| Pérdidas sistema | 14.0%          |
| Costo sistema    | 1998.0 €       |
| Interés anual    | 5.0%           |
| Vida útil        | 20 años        |

Tabla 5: (G3) Datos proporcionados

| Parámetro                   | Valor                      |
|-----------------------------|----------------------------|
| Ángulo de inclinación       | 26°                        |
| Ángulo de azimut            | 0°                         |
| Producción diaria (E_d)     | 9.04 kWh                   |
| Producción mensual (E_m)    | 275.0 kWh                  |
| Producción anual (E_y)      | 3300.01 kWh                |
| Irradiación diaria (H_d)    | 5.86 kWh/m <sup>2</sup>    |
| Irradiación mensual (H_m)   | 178.33 kWh/m <sup>2</sup>  |
| Irradiación anual (H_y)     | 2139.95 kWh/m <sup>2</sup> |
| Desv. estándar mensual      | 6.91 kWh                   |
| Desv. estándar anual        | 82.94 kWh                  |
| Pérdida ángulo incidencia   | -2.65%                     |
| Pérdida espectral           | ?(0)                       |
| Pérdida temp. e irradiancia | -7.9%                      |
| Pérdidas totales            | -22.9%                     |
| LCOE                        | 0.06 €/kWh                 |

Tabla 6: (G3) Resultados de la simulación

| month | E_d   | E_m    | SD_m  |
|-------|-------|--------|-------|
| 1     | 7.05  | 218.63 | 23.66 |
| 2     | 7.99  | 223.73 | 25.75 |
| 3     | 8.83  | 273.62 | 36.06 |
| 4     | 9.75  | 292.57 | 24.03 |
| 5     | 10.62 | 329.25 | 22.39 |
| 6     | 11.16 | 334.83 | 10.9  |
| 7     | 11.19 | 346.86 | 7.06  |
| 8     | 10.72 | 332.17 | 10.08 |
| 9     | 9.63  | 288.78 | 14.6  |
| 10    | 8.09  | 250.81 | 22.29 |
| 11    | 6.95  | 208.61 | 21.7  |
| 12    | 6.46  | 200.14 | 20.13 |

Tabla 7: (G3) Energía FV mensual

3. Grupo FV G4.



Figura 9: (G4) Paneles FV.

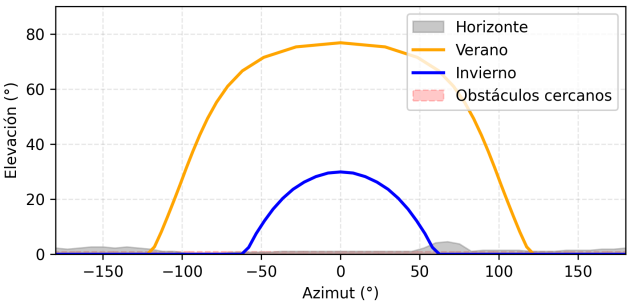


Figura 10: (G4) Perfil del horizonte y sombras.

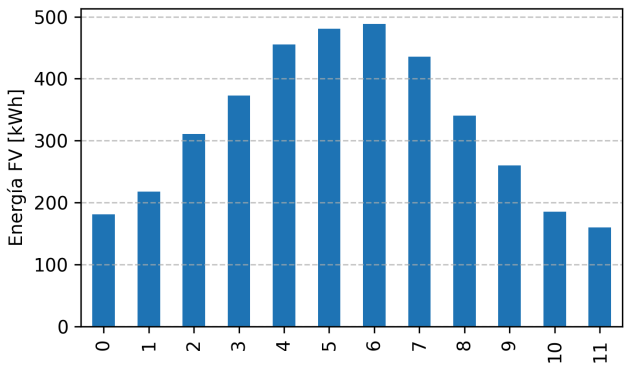


Figura 11: (G4) Producción de energía mensual.

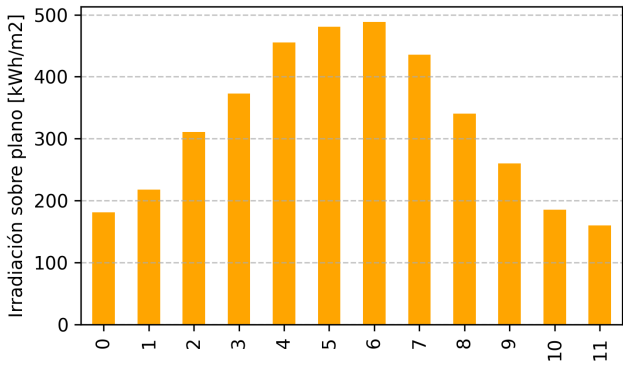


Figura 12: (G4) Irradiación mensual sobre plano fijo.

| Parámetro        | Valor          |
|------------------|----------------|
| Latitud/Longitud | 36.667, -4.461 |
| Horizonte        | DEM-calculated |
| Base de datos    | PVGIS-SARAH3   |
| Tecnología FV    | c-Si           |
| FV instalado     | 2.7 kWp        |
| Pérdidas sistema | 14.0%          |
| Costo sistema    | 2700.0 €       |
| Interés anual    | 5.0%           |
| Vida útil        | 20 años        |

Tabla 8: (G4) Datos proporcionados

| Parámetro                   | Valor                      |
|-----------------------------|----------------------------|
| Ángulo de inclinación       | 0°                         |
| Ángulo de azimut            |                            |
| Producción diaria (E_d)     | 10.65 kWh                  |
| Producción mensual (E_m)    | 324.03 kWh                 |
| Producción anual (E_y)      | 3888.3 kWh                 |
| Irradiación diaria (H_d)    | 5.15 kWh/m <sup>2</sup>    |
| Irradiación mensual (H_m)   | 156.72 kWh/m <sup>2</sup>  |
| Irradiación anual (H_y)     | 1880.65 kWh/m <sup>2</sup> |
| Desv. estándar mensual      | 6.34 kWh                   |
| Desv. estándar anual        | 76.05 kWh                  |
| Pérdida ángulo incidencia   | -3.51%                     |
| Pérdida espectral           | ?(0)                       |
| Pérdida temp. e irradiancia | -7.72%                     |
| Pérdidas totales            | -23.42%                    |
| LCOE                        | 0.07 €/kWh                 |

Tabla 9: (G4) Resultados de la simulación

| month | E_d   | E_m    | SD_m  |
|-------|-------|--------|-------|
| 1     | 5.84  | 181.03 | 16.25 |
| 2     | 7.78  | 217.98 | 20.6  |
| 3     | 10.01 | 310.45 | 35.76 |
| 4     | 12.43 | 372.8  | 28.3  |
| 5     | 14.69 | 455.28 | 30.74 |
| 6     | 16.03 | 480.98 | 15.75 |
| 7     | 15.77 | 488.76 | 10.1  |
| 8     | 14.04 | 435.38 | 12.81 |
| 9     | 11.35 | 340.45 | 14.88 |
| 10    | 8.39  | 259.99 | 18.92 |
| 11    | 6.18  | 185.39 | 15.98 |
| 12    | 5.16  | 159.81 | 12.9  |

Tabla 10: (G4) Energía FV mensual

4. Grupo FV G5.



Figura 13: (G5) Paneles FV.

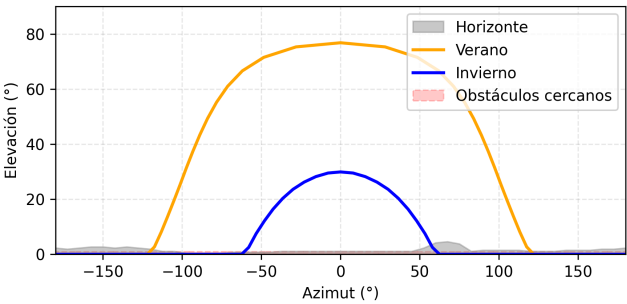


Figura 14: (G5) Perfil del horizonte y sombras.

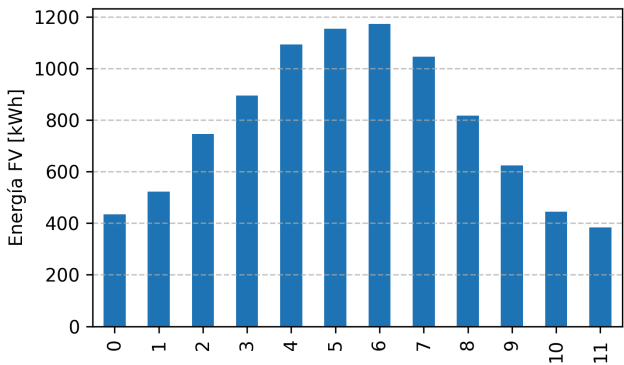


Figura 15: (G5) Producción de energía mensual.

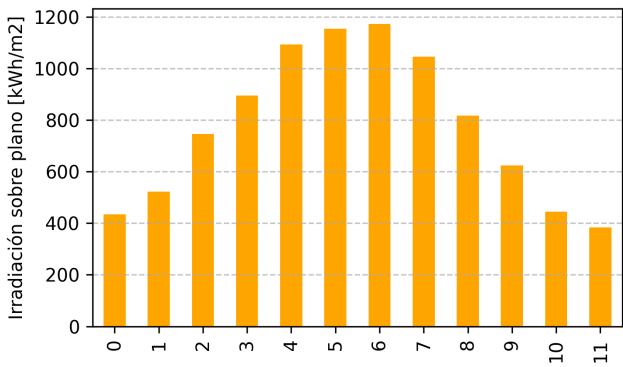


Figura 16: (G5) Irradiación mensual sobre plano fijo.

| Parámetro        | Valor          |
|------------------|----------------|
| Latitud/Longitud | 36.667, -4.461 |
| Horizonte        | DEM-calculated |
| Base de datos    | PVGIS-SARAH3   |
| Tecnología FV    | c-Si           |
| FV instalado     | 6.48 kWp       |
| Pérdidas sistema | 14.0%          |
| Costo sistema    | 6480.0 €       |
| Interés anual    | 5.0%           |
| Vida útil        | 20 años        |

Tabla 11: (G5) Datos proporcionados

| Parámetro                   | Valor                      |
|-----------------------------|----------------------------|
| Ángulo de inclinación       | 0°                         |
| Ángulo de azimut            |                            |
| Producción diaria (E_d)     | 25.57 kWh                  |
| Producción mensual (E_m)    | 777.66 kWh                 |
| Producción anual (E_y)      | 9331.92 kWh                |
| Irradiación diaria (H_d)    | 5.15 kWh/m <sup>2</sup>    |
| Irradiación mensual (H_m)   | 156.72 kWh/m <sup>2</sup>  |
| Irradiación anual (H_y)     | 1880.65 kWh/m <sup>2</sup> |
| Desv. estándar mensual      | 15.21 kWh                  |
| Desv. estándar anual        | 182.52 kWh                 |
| Pérdida ángulo incidencia   | -3.51%                     |
| Pérdida espectral           | ?(0)                       |
| Pérdida temp. e irradiancia | -7.72%                     |
| Pérdidas totales            | -23.42%                    |
| LCOE                        | 0.07 €/kWh                 |

Tabla 12: (G5) Resultados de la simulación

| month | E_d   | E_m     | SD_m  |
|-------|-------|---------|-------|
| 1     | 14.02 | 434.47  | 39.0  |
| 2     | 18.68 | 523.14  | 49.43 |
| 3     | 24.04 | 745.09  | 85.84 |
| 4     | 29.82 | 894.72  | 67.93 |
| 5     | 35.25 | 1092.66 | 73.77 |
| 6     | 38.48 | 1154.35 | 37.79 |
| 7     | 37.84 | 1173.03 | 24.25 |
| 8     | 33.71 | 1044.92 | 30.75 |
| 9     | 27.24 | 817.08  | 35.72 |
| 10    | 20.13 | 623.99  | 45.41 |
| 11    | 14.83 | 444.95  | 38.36 |
| 12    | 12.37 | 383.53  | 30.95 |

Tabla 13: (G5) Energía FV mensual

5. Grupo FV Total.



Figura 17: (Total) Paneles FV.

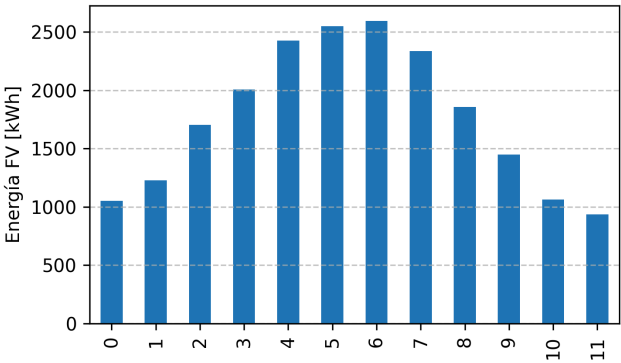


Figura 18: (Total) Producción de energía mensual.

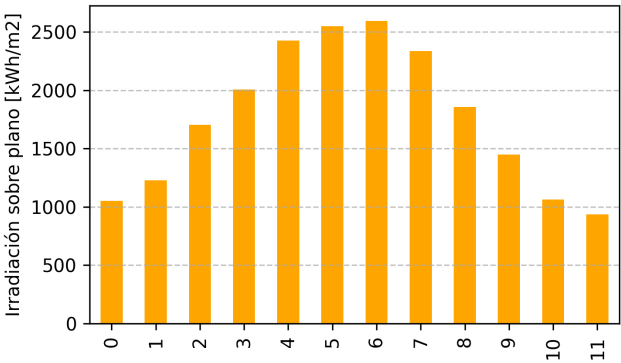


Figura 19: (Total) Irradiación mensual sobre plano fijo.

| Parámetro        | Valor          |
|------------------|----------------|
| Latitud/Longitud | 36.67, -4.46   |
| Horizonte        | DEM-calculated |
| Base de datos    | PVGIS-SARAH3   |
| Tecnología FV    | c-Si           |
| FV instalado     | 14.42 kWp      |
| Pérdidas sistema | 14.0%          |
| Costo sistema    | 3240.0 €       |
| Interés anual    | 5.0%           |
| Vida útil        | 20 años        |

Tabla 14: (Total) Datos proporcionados

| Parámetro                   | Valor          |
|-----------------------------|----------------|
| Ángulo de inclinación       | 0°             |
| Ángulo de azimut            |                |
| Producción diaria (E_d)     | 58.04 kWh      |
| Producción mensual (E_m)    | 1765.52 kWh    |
| Producción anual (E_y)      | 21186.19 kWh   |
| Irradiación diaria (H_d)    | 5.25 kWh/m²    |
| Irradiación mensual (H_m)   | 159.72 kWh/m²  |
| Irradiación anual (H_y)     | 1916.61 kWh/m² |
| Desv. estándar mensual      | 36.07 kWh      |
| Desv. estándar anual        | 432.77 kWh     |
| Pérdida ángulo incidencia   | -3.39%         |
| Pérdida espectral           | 0.0            |
| Pérdida temp. e irradiancia | -7.74%         |
| Pérdidas totales            | -23.35%        |
| LCOE                        | 0.07 €/kWh     |

Tabla 15: (Total) Resultados de la simulación

| month | E_d   | E_m     | SD_m   |
|-------|-------|---------|--------|
| 1     | 33.92 | 1051.37 | 98.41  |
| 2     | 43.79 | 1226.42 | 120.5  |
| 3     | 54.9  | 1701.7  | 200.58 |
| 4     | 66.91 | 2007.45 | 154.22 |
| 5     | 78.18 | 2423.52 | 163.78 |
| 6     | 84.91 | 2547.34 | 83.33  |
| 7     | 83.72 | 2595.16 | 53.54  |
| 8     | 75.32 | 2334.93 | 69.02  |
| 9     | 61.84 | 1854.85 | 83.06  |
| 10    | 46.67 | 1446.78 | 109.33 |
| 11    | 35.38 | 1061.42 | 95.22  |
| 12    | 30.18 | 935.25  | 79.46  |

Tabla 16: (Total) Energía FV mensual

## **6. Presupuesto**

| Capítulo                   | Categoría                | Variable                              | Fijo (€) | €/Wdc | MO (%) | Mat    | MO     | Total  |
|----------------------------|--------------------------|---------------------------------------|----------|-------|--------|--------|--------|--------|
| Coste Directo              | Equipos Principales      | Paneles fotovoltaicos monocristalinos | 0.0      | 0.2   | 5.0    | 0.2    | 0.0    | 0.2    |
|                            |                          | Inversor string/híbrido               | 150.0    | 0.1   | 5.0    | 142.6  | 7.5    | 150.1  |
|                            |                          | Sistema de almacenamiento (Batería)   | 3000.0   | 0.2   | 5.0    | 2850.1 | 150.0  | 3000.1 |
|                            | Balance de Sistema (BOS) | Soportes, cableado y protecciones     | 750.0    | 0.2   | 40.0   | 450.1  | 300.1  | 750.2  |
|                            |                          | Sistema de monitorización             | 150.0    | 0.0   | 10.0   | 135.0  | 15.0   | 150.0  |
|                            | Obra Civil               | Adecuación y pequeñas adaptaciones    | 200.0    | 0.0   | 80.0   | 40.0   | 160.0  | 200.0  |
| Coste Directo - Subtotal   |                          |                                       | 4250.0   | 0.7   | 145.0  | 3618.0 | 632.6  | 4250.6 |
| Coste Indirecto            | Ingeniería               | Proyecto, dirección y legalización    | 900.0    | 0.0   | 100.0  | 0.0    | 900.0  | 900.0  |
|                            | Interconexión            | Trámites con distribuidora y tasas    | 500.0    | 0.0   | 90.0   | 50.0   | 450.0  | 500.0  |
|                            | Gastos Generales         | Margen comercial y overhead           | 0.0      | 0.2   | 0.0    | 0.2    | 0.0    | 0.2    |
|                            | Contingencias            | Reserva para imprevistos (3-5%)       | 0.0      | 0.0   | 0.0    | 0.0    | 0.0    | 0.0    |
| Coste Indirecto - Subtotal |                          |                                       | 1400.0   | 0.2   | 190.0  | 50.2   | 1350.0 | 1400.2 |
| TOTAL GENERAL              |                          |                                       | 5650.0   | 1.0   |        | 3668.2 | 1982.6 | 5650.8 |

Tabla 17: Presupuesto

7. inputs

| Categoría    | Variable   | valor                   |
|--------------|------------|-------------------------|
| Proyecto     | nombre     | Mi rrrr Real            |
| Autor        |            | P. Román                |
|              | N Registro | N.12345                 |
| Organización | nombre     | Instalaciones S..       |
|              | CIF        | A12345678               |
|              | dirección  | Calle Iglesia, 123      |
|              | teléfono   | 912 34 56 78            |
|              | correo     | pperezinstalaciones.com |
| Destinatario | nombre     | Ayto. Villanueva        |
|              | CIF        |                         |
|              | dirección  | Plaza Mayor, 1          |

Tabla 18: PDF



| Capítulo    | Categoría            | Subcategoría             | Consumo_diario_kWh | Perfil_24h                                                                                                                                                                                                               |
|-------------|----------------------|--------------------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Residencial | Vivienda Estándar    | Uso General              | 10                 | [0.024, 0.016, 0.016, 0.016, 0.016, 0.024, 0.032, 0.145, 0.04, 0.032, 0.024, 0.024, 0.032, 0.04, 0.032, 0.024, 0.032, 0.047, 0.063, 0.08, 0.095, 0.071, 0.047, 0.032]                                                    |
|             | Vivienda Unifamiliar | Aire Acondicionado       | 12                 | [0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.02, 0.05, 0.08, 0.12, 0.15, 0.15, 0.12, 0.1, 0.08, 0.06, 0.04, 0.02, 0.01, 0.0, 0.0, 0.0]                                                                                     |
|             |                      | Calefacción (B. Calor)   | 20                 | [0.05, 0.06, 0.07, 0.07, 0.07, 0.08, 0.1, 0.12, 0.08, 0.05, 0.03, 0.02, 0.02, 0.02, 0.02, 0.02, 0.02, 0.03, 0.04, 0.05, 0.06, 0.06, 0.04, 0.04]                                                                          |
|             |                      | Cargador Coche Eléctrico | 15                 | [0.12, 0.12, 0.12, 0.12, 0.12, 0.12, 0.12, 0.12, 0.04, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.04]                                                                                            |
| Terciario   | Bloque Pisos         | Servicios Comunes        | 30                 | [0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416] |
|             | Oficina Moderna      | Iluminación LED          | 15                 | [0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.05, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.05, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0]                                                                                          |
|             |                      | Climatización            | 50                 | [0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.05, 0.1, 0.12, 0.12, 0.12, 0.12, 0.05, 0.12, 0.12, 0.06, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0]                                                                                        |
|             | Comercio             | Rótulos Luminosos        | 5                  | [0.1, 0.1, 0.1, 0.1, 0.1, 0.05, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.05, 0.1, 0.1, 0.1, 0.1]                                                                                                    |
| Industrial  | Nave Producción      | Maquinaria Pesada        | 350                | [0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.05, 0.15, 0.15, 0.15, 0.15, 0.15, 0.05, 0.15, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0]                                                                                              |
|             |                      | Aire Comprimido          | 40                 | [0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416] |
|             | Almacén Logístico    | Carga Carretillas        | 60                 | [0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.1, 0.1, 0.1, 0.1]                                                                                                 |
| Agrario     | Explotación Agrícola | Bombeo Riego             | 150                | [0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.1, 0.15, 0.15, 0.15, 0.15, 0.15, 0.15, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0]                                                                                           |
|             | Granja Ganadera      | Ventilación Forzada      | 25                 | [0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416, 0.0416] |
|             |                      | Ordeño Automático        | 15                 | [0.0, 0.0, 0.0, 0.0, 0.0, 0.2, 0.2, 0.1, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.2, 0.2, 0.1, 0.0, 0.0, 0.0, 0.0]                                                                                                      |

Tabla 20: CONSUMO

| Capítulo        | Categoría           | Variable                              | Fijo (€) | €/Wdc | MO (%) |
|-----------------|---------------------|---------------------------------------|----------|-------|--------|
| Coste Directo   | Equipos Principales | Paneles fotovoltaicos monocristalinos | 0        | 0.22  | 5      |
|                 |                     | Inversor string/híbrido               | 150      | 0.12  | 5      |
|                 |                     | Sistema de almacenamiento (Batería)   | 3000     | 0.15  | 5      |
|                 |                     | Balance de Sistema (BOS)              | 750      | 0.19  | 40     |
|                 |                     | Sistema de monitorización             | 150      | 0.02  | 10     |
|                 | Obra Civil          | Adecuación y pequeñas adaptaciones    | 200      | 0.005 | 80     |
| Coste Indirecto | Ingeniería          | Proyecto, dirección y legalización    | 900      | 0.02  | 100    |
|                 | Interconexión       | Trámites con distribuidora y tasas    | 500      | 0.01  | 90     |
|                 | Gastos Generales    | Margen comercial y overhead           | 0        | 0.18  | 0      |
|                 | Contingencias       | Reserva para imprevistos (3-5%)       | 0        | 0.04  | 0      |
| Otros           | Otros               |                                       | 0        | 0.0   | 0      |

Tabla 21: COSTES DE INSTALACION

| Capítulo        | Categoría            | Concepto                    | Año_inicio | Año_fin | Fijo (€/año) | Variable | Unidad_variable  |
|-----------------|----------------------|-----------------------------|------------|---------|--------------|----------|------------------|
| Operación FV    | Energía fotovoltaica | PV_Fijo_Anual               | 1          | 25      | 0.0          | 0        | €/año            |
|                 |                      | PV_Fijo_por_Capacidad       | 1          | 25      | 19.0         | 0        | €/Wdc            |
|                 |                      | PV_Variable_Generacion      | 1          | 25      | 0.0          | 0        | €/kWh            |
| Batería         | Almacenamiento       | Bateria_Fijo_Anual          | 1          | 15      | 0.0          | 0        | €/año            |
|                 |                      | Bateria_Fijo_por_Capacidad  | 1          | 15      | 7.25         | 0        | €/kWh_instalada  |
|                 |                      | Bateria_Variable_Descarga   | 1          | 15      | 0.0          | 0        | €/kWh_descargado |
|                 | Reemplazo de activos | Bateria_Coste_Reemplazo     | 10         | 10      | 252.0        | 0        | €/kWh_instalada  |
| Infraestructura | Uso de suelo         | Arrendamiento_Anual_Terreno | 1          | 25      | 0.0          | 0        | €/año            |
| Otrsos          | Otdros               | Otros                       | 1          | 25      | 0.0          | 0        | €                |

Tabla 22: COSTES DE OPERACION

| Capítulo                | Categoría               | Concepto                  | Año_inicio | Año_fin | Ingreso_fijo (€/año) | Ingreso_variable | Unidad_variable     |
|-------------------------|-------------------------|---------------------------|------------|---------|----------------------|------------------|---------------------|
| Ingresos de explotación | Autoconsumo             | Ahorro autoconsumo        | 1          | 25      | 0                    | 0.18             | €/kWh autoconsumido |
|                         | Excedentes              | Venta excedentes          | 1          | 25      | 0                    | 0.08             | €/kWh exportado     |
|                         |                         | Compensación simplificada | 1          | 25      | 0                    | 0.06             | €/kWh compensado    |
| Ingresos regulados      | Certificados            | CAE                       | 1          | 10      | 0                    | 0.14             | €/kWh generado      |
|                         |                         | Garantías de origen       | 1          | 25      | 0                    | 0.005            | €/kWh generado      |
| Subvenciones            | Bonificaciones fiscales | Bonificación IBI          | 1          | 5       | 100                  | 0.0              | €/año               |
|                         | Deducciones fiscales    | Deducción IRPF            | 1          | 1       | 200                  | 0.0              | €/año               |
|                         | Inversión               | Next Generation           | 1          | 1       | 0                    | 0.3              | % CAPEX             |
|                         | Locales/autonómicas     | ICIO                      | 1          | 1       | 100                  | 0.0              | €/año               |
| Otros                   | Otros                   | Otros                     | 1          | 25      | 0                    | 0.0              | €                   |

Tabla 23: INGRESOS

| Capítulo            | Categoría    | Parámetro                | Valor  | Unidad           |
|---------------------|--------------|--------------------------|--------|------------------|
| Análisis Financiero | Proyecto     | Duración de la actuación | 2033.0 | años             |
|                     |              | Impuestos (IVA)          | 21.0   | %                |
|                     | Mercado      | Tasa de inflación        | 2.5    | %/año            |
|                     |              | Precio de la energía     | 0.12   | €/kWh            |
|                     |              | Tasa de descuento (VAN)  | 0.08   | -                |
|                     | Financiación | Porcentaje financiado    | 27.0   | %                |
|                     |              | Años de financiación     | 10.0   | años             |
|                     |              | Tipo de interés anual    | 7.0    | %                |
|                     | Operación    | Tasa seguro anual        | 0.0    | % de instalación |

Tabla 24: FINANCIEROS

## 8. printhorizon

### 8.1. printhorizon

- **inputs:**
  - **location:**
    - **latitude:** 36.666579105549324
    - **longitude:** -4.4610378208142265
    - **elevation:** 3.0
  - **horizon\_db:** DEM-calculated
- **outputs:**
  - **horizon\_profile:**
    - **A:** -180.0
    - **H\_hor:** 2.3
    - **A:** -172.5
    - **H\_hor:** 1.9
    - **A:** -165.0
    - **H\_hor:** 2.3
    - **A:** -157.5
    - **H\_hor:** 2.7
    - **A:** -150.0
    - **H\_hor:** 2.7
    - **A:** -142.5
    - **H\_hor:** 2.3
    - **A:** -135.0
    - **H\_hor:** 2.7
    - **A:** -127.5
    - **H\_hor:** 2.3
    - **A:** -120.0
    - **H\_hor:** 1.9
    - **A:** -112.5
    - **H\_hor:** 1.1
    - **A:** -105.0
    - **H\_hor:** 1.1
    - **A:** -97.5
    - **H\_hor:** 0.4
    - **A:** -90.0
    - **H\_hor:** 0.4
    - **A:** -82.5
    - **H\_hor:** 0.4
    - **A:** -75.0
    - **H\_hor:** 0.4
    - **A:** -67.5
    - **H\_hor:** 0.8
    - **A:** -60.0
    - **H\_hor:** 0.8

- **A:** -52.5
- **H\_hor:** 0.8
- **A:** -45.0
- **H\_hor:** 0.8
- **A:** -37.5
- **H\_hor:** 1.1
- **A:** -30.0
- **H\_hor:** 1.1
- **A:** -22.5
- **H\_hor:** 1.1
- **A:** -15.0
- **H\_hor:** 1.1
- **A:** -7.5
- **H\_hor:** 1.1
- **A:** 0.0
- **H\_hor:** 1.1
- **A:** 7.5
- **H\_hor:** 1.1
- **A:** 15.0
- **H\_hor:** 1.1
- **A:** 22.5
- **H\_hor:** 1.1
- **A:** 30.0
- **H\_hor:** 1.1
- **A:** 37.5
- **H\_hor:** 1.1
- **A:** 45.0
- **H\_hor:** 1.1
- **A:** 52.5
- **H\_hor:** 2.3
- **A:** 60.0
- **H\_hor:** 4.2
- **A:** 67.5
- **H\_hor:** 4.6
- **A:** 75.0
- **H\_hor:** 3.8
- **A:** 82.5
- **H\_hor:** 1.1
- **A:** 90.0
- **H\_hor:** 1.5
- **A:** 97.5
- **H\_hor:** 1.5
- **A:** 105.0
- **H\_hor:** 1.5
- **A:** 112.5
- **H\_hor:** 1.5
- **A:** 120.0

- **H\_hor:** 1.1
- **A:** 127.5
- **H\_hor:** 1.1
- **A:** 135.0
- **H\_hor:** 1.1
- **A:** 142.5
- **H\_hor:** 1.5
- **A:** 150.0
- **H\_hor:** 1.5
- **A:** 157.5
- **H\_hor:** 1.5
- **A:** 165.0
- **H\_hor:** 1.9
- **A:** 172.5
- **H\_hor:** 1.9
- **A:** 180.0
- **H\_hor:** 2.3
- **winter\_solstice:**
  - **A\_sun(w):** -180.0
  - **H\_sun(w):** 0.0
  - **A\_sun(w):** -151.9
  - **H\_sun(w):** 0.0
  - **A\_sun(w):** -131.5
  - **H\_sun(w):** 0.0
  - **A\_sun(w):** -118.0
  - **H\_sun(w):** 0.0
  - **A\_sun(w):** -108.7
  - **H\_sun(w):** 0.0
  - **A\_sun(w):** -101.7
  - **H\_sun(w):** 0.0
  - **A\_sun(w):** -96.0
  - **H\_sun(w):** 0.0
  - **A\_sun(w):** -91.1
  - **H\_sun(w):** 0.0
  - **A\_sun(w):** -86.7
  - **H\_sun(w):** 0.0
  - **A\_sun(w):** -82.6
  - **H\_sun(w):** 0.0
  - **A\_sun(w):** -78.7
  - **H\_sun(w):** 0.0
  - **A\_sun(w):** -74.8
  - **H\_sun(w):** 0.0
  - **A\_sun(w):** -70.8
  - **H\_sun(w):** 0.0
  - **A\_sun(w):** -66.8
  - **H\_sun(w):** 0.0
  - **A\_sun(w):** -62.5

- H\_sun(w): 0.0  
 - A\_sun(w): -58.0  
 - H\_sun(w): 2.5  
 - A\_sun(w): -53.3  
 - H\_sun(w): 7.5  
 - A\_sun(w): -48.1  
 - H\_sun(w): 12.1  
 - A\_sun(w): -42.6  
 - H\_sun(w): 16.4  
 - A\_sun(w): -36.5  
 - H\_sun(w): 20.3  
 - A\_sun(w): -30.0  
 - H\_sun(w): 23.6  
 - A\_sun(w): -23.0  
 - H\_sun(w): 26.2  
 - A\_sun(w): -15.6  
 - H\_sun(w): 28.2  
 - A\_sun(w): -7.9  
 - H\_sun(w): 29.5  
 - A\_sun(w): 0.0  
 - H\_sun(w): 29.9  
 - A\_sun(w): 7.9  
 - H\_sun(w): 29.5  
 - A\_sun(w): 15.6  
 - H\_sun(w): 28.2  
 - A\_sun(w): 23.0  
 - H\_sun(w): 26.2  
 - A\_sun(w): 30.0  
 - H\_sun(w): 23.6  
 - A\_sun(w): 36.5  
 - H\_sun(w): 20.3  
 - A\_sun(w): 42.6  
 - H\_sun(w): 16.4  
 - A\_sun(w): 48.1  
 - H\_sun(w): 12.1  
 - A\_sun(w): 53.3  
 - H\_sun(w): 7.5  
 - A\_sun(w): 58.0  
 - H\_sun(w): 2.5  
 - A\_sun(w): 62.5  
 - H\_sun(w): 0.0  
 - A\_sun(w): 66.8  
 - H\_sun(w): 0.0  
 - A\_sun(w): 70.8  
 - H\_sun(w): 0.0  
 - A\_sun(w): 74.8  
 - H\_sun(w): 0.0

- A\_sun(w): 78.7
- H\_sun(w): 0.0
- A\_sun(w): 82.6
- H\_sun(w): 0.0
- A\_sun(w): 86.7
- H\_sun(w): 0.0
- A\_sun(w): 91.1
- H\_sun(w): 0.0
- A\_sun(w): 96.0
- H\_sun(w): 0.0
- A\_sun(w): 101.7
- H\_sun(w): 0.0
- A\_sun(w): 108.7
- H\_sun(w): 0.0
- A\_sun(w): 118.0
- H\_sun(w): 0.0
- A\_sun(w): 131.5
- H\_sun(w): 0.0
- A\_sun(w): 151.9
- H\_sun(w): 0.0
- A\_sun(w): 180.0
- H\_sun(w): 0.0
- summer\_solstice:
- A\_sun(s): -180.0
- H\_sun(s): 0.0
- A\_sun(s): -172.1
- H\_sun(s): 0.0
- A\_sun(s): -164.4
- H\_sun(s): 0.0
- A\_sun(s): -157.0
- H\_sun(s): 0.0
- A\_sun(s): -150.0
- H\_sun(s): 0.0
- A\_sun(s): -143.5
- H\_sun(s): 0.0
- A\_sun(s): -137.4
- H\_sun(s): 0.0
- A\_sun(s): -131.9
- H\_sun(s): 0.0
- A\_sun(s): -126.7
- H\_sun(s): 0.0
- A\_sun(s): -122.0
- H\_sun(s): 0.0
- A\_sun(s): -117.5
- H\_sun(s): 2.7
- A\_sun(s): -113.2
- H\_sun(s): 8.1

- A\_sun(s): -109.2  
 - H\_sun(s): 13.7  
 - A\_sun(s): -105.2  
 - H\_sun(s): 19.5  
 - A\_sun(s): -101.3  
 - H\_sun(s): 25.3  
 - A\_sun(s): -97.4  
 - H\_sun(s): 31.3  
 - A\_sun(s): -93.3  
 - H\_sun(s): 37.3  
 - A\_sun(s): -88.9  
 - H\_sun(s): 43.3  
 - A\_sun(s): -84.0  
 - H\_sun(s): 49.3  
 - A\_sun(s): -78.3  
 - H\_sun(s): 55.2  
 - A\_sun(s): -71.3  
 - H\_sun(s): 61.0  
 - A\_sun(s): -62.0  
 - H\_sun(s): 66.6  
 - A\_sun(s): -48.5  
 - H\_sun(s): 71.5  
 - A\_sun(s): -28.1  
 - H\_sun(s): 75.3  
 - A\_sun(s): 0.0  
 - H\_sun(s): 76.8  
 - A\_sun(s): 28.1  
 - H\_sun(s): 75.3  
 - A\_sun(s): 48.5  
 - H\_sun(s): 71.5  
 - A\_sun(s): 62.0  
 - H\_sun(s): 66.6  
 - A\_sun(s): 71.3  
 - H\_sun(s): 61.0  
 - A\_sun(s): 78.3  
 - H\_sun(s): 55.2  
 - A\_sun(s): 84.0  
 - H\_sun(s): 49.3  
 - A\_sun(s): 88.9  
 - H\_sun(s): 43.3  
 - A\_sun(s): 93.3  
 - H\_sun(s): 37.3  
 - A\_sun(s): 97.4  
 - H\_sun(s): 31.3  
 - A\_sun(s): 101.3  
 - H\_sun(s): 25.3  
 - A\_sun(s): 105.2

- **H\_sun(s):** 19.5
- **A\_sun(s):** 109.2
- **H\_sun(s):** 13.7
- **A\_sun(s):** 113.2
- **H\_sun(s):** 8.1
- **A\_sun(s):** 117.5
- **H\_sun(s):** 2.7
- **A\_sun(s):** 122.0
- **H\_sun(s):** 0.0
- **A\_sun(s):** 126.7
- **H\_sun(s):** 0.0
- **A\_sun(s):** 131.9
- **H\_sun(s):** 0.0
- **A\_sun(s):** 137.4
- **H\_sun(s):** 0.0
- **A\_sun(s):** 143.5
- **H\_sun(s):** 0.0
- **A\_sun(s):** 150.0
- **H\_sun(s):** 0.0
- **A\_sun(s):** 157.0
- **H\_sun(s):** 0.0
- **A\_sun(s):** 164.4
- **H\_sun(s):** 0.0
- **A\_sun(s):** 172.1
- **H\_sun(s):** 0.0
- **A\_sun(s):** 180.0
- **H\_sun(s):** 0.0
- **meta:**
  - **inputs:**
    - **location:**
      - **description:** Selected location
      - **variables:**
        - **latitude:**
          - **description:** Latitude
          - **units:** decimal degree
        - **longitude:**
          - **description:** Longitude
          - **units:** decimal degree
        - **elevation:**
          - **description:** Elevation
          - **units:** m
    - **horizon\_db:**
      - **description:** Source of horizon data
    - **outputs:**
      - **horizon\_profile:**
        - **type:** series
        - **description:** Horizon profile
  - **variables:**
    - **A:**
      - **description:** Azimuth (0 = S, 90 = W, -90 = E)
      - **units:** degree
    - **H\_hor:**
      - **description:** Horizon height
      - **units:** degree
  - **winter\_solstice:**
    - **type:** series
    - **description:** Winter solstice
    - **variables:**
      - **A\_sun(w):**
        - **description:** Sun azimuth in the winter solstice (Dec 21) (0 = S, 90 = W, -90 = E)
        - **units:** degree
      - **H\_sun(w):**
        - **description:** Sun height in the winter solstice (Dec 21)
        - **units:** degree
  - **summer\_solstice:**
    - **type:** series
    - **description:** Summer solstice profile
    - **variables:**
      - **A\_sun(s):**
        - **description:** Sun azimuth in the summer solstice (June 21) (0 = S, 90 = W, -90 = E)
        - **units:** degree
      - **H\_sun(s):**
        - **description:** Sun height in the summer solstice (June 21)
        - **units:** degree



## 9. printhorizon

### 9.1. G5

- **inputs:**
  - **location:**
    - **latitude:** 36.666579105549324
    - **longitude:** −4.4610378208142265
    - **elevation:** 3.0
  - **meteo\_data:**
    - **radiation\_db:** PVGIS-SARAH3
    - **meteo\_db:** ERA5
    - **year\_min:** 2005
    - **year\_max:** 2023
    - **use\_horizon:** True
    - **horizon\_db:** DEM-calculated
  - **mounting\_system:**
    - **fixed:**
      - **slope:**
        - **value:** 0
        - **optimal:** False
      - **azimuth:**
        - **value:** 0
        - **optimal:** False
      - **type:** free-standing
    - **pv\_module:**
      - **technology:** c-Si
      - **peak\_power:** 6.48
      - **system\_loss:** 14.0
    - **economic\_data:**
      - **system\_cost:** 6480.0
      - **interest:** 5.0
      - **lifetime:** 20
  - **outputs:**
    - **monthly:**
      - **fixed:**
        - **month:** 1
        - **E\_d:** 14.02
        - **E\_m:** 434.47
        - **H(i)\_d:** 2.78
        - **H(i)\_m:** 86.25
        - **SD\_m:** 39.0
        - **month:** 2
        - **E\_d:** 18.68
        - **E\_m:** 523.14
        - **H(i)\_d:** 3.63
        - **H(i)\_m:** 101.69
        - **SD\_m:** 49.43
        - **month:** 3
        - **E\_d:** 24.04
        - **E\_m:** 745.09
        - **H(i)\_d:** 4.68
        - **H(i)\_m:** 145.12
        - **SD\_m:** 85.84
        - **month:** 4
        - **E\_d:** 29.82
        - **E\_m:** 894.72
        - **H(i)\_d:** 5.87
        - **H(i)\_m:** 176.25
        - **SD\_m:** 67.93
        - **month:** 5
        - **E\_d:** 35.25
        - **E\_m:** 1092.66
        - **H(i)\_d:** 7.06
        - **H(i)\_m:** 218.93
        - **SD\_m:** 73.77
        - **month:** 6
        - **E\_d:** 38.48
        - **E\_m:** 1154.35
        - **H(i)\_d:** 7.85
        - **H(i)\_m:** 235.42
        - **SD\_m:** 37.79
        - **month:** 7
        - **E\_d:** 37.84
        - **E\_m:** 1173.03
        - **H(i)\_d:** 7.84
        - **H(i)\_m:** 243.17
        - **SD\_m:** 24.25
        - **month:** 8
        - **E\_d:** 33.71
        - **E\_m:** 1044.92
        - **H(i)\_d:** 6.97
        - **H(i)\_m:** 216.12
        - **SD\_m:** 30.75
        - **month:** 9
        - **E\_d:** 27.24
        - **E\_m:** 817.08
        - **H(i)\_d:** 5.54
        - **H(i)\_m:** 166.16
        - **SD\_m:** 35.72
        - **month:** 10
        - **E\_d:** 20.13
        - **E\_m:** 623.99
        - **H(i)\_d:** 4.06
        - **H(i)\_m:** 125.83
        - **SD\_m:** 45.41
        - **month:** 11
        - **E\_d:** 14.83
        - **E\_m:** 444.95
        - **H(i)\_d:** 2.96
        - **H(i)\_m:** 88.76
        - **SD\_m:** 38.36
        - **month:** 12
        - **E\_d:** 12.37
        - **E\_m:** 383.53
        - **H(i)\_d:** 2.48
        - **H(i)\_m:** 76.97
        - **SD\_m:** 30.95
      - **totals:**
        - **fixed:**
          - **E\_d:** 25.57
          - **E\_m:** 777.66
          - **E\_y:** 9331.92
          - **H(i)\_d:** 5.15
          - **H(i)\_m:** 156.72
          - **H(i)\_y:** 1880.65
          - **SD\_m:** 15.21
          - **SD\_y:** 182.52
          - **l\_aoi:** −3.51
          - **l\_spec:** ?(0)
          - **l\_tg:** −7.72
          - **l\_total:** −23.42
          - **LCOE\_pv:** 0.07
    - **meta:**
      - **inputs:**
        - **location:**
          - **description:** Selected location
        - **variables:**
          - **latitude:**
            - **description:** Latitude
            - **units:** decimal degree
          - **longitude:**
            - **description:** Longitude
            - **units:** decimal degree
          - **elevation:**
            - **description:** Elevation
            - **units:** m
        - **meteo\_data:**
          - **description:** Sources of meteorological data
          - **variables:**

- ▶ **radiation\_db:**
  - **description:** Solar radiation database
- ▶ **meteo\_db:**
  - **description:** Database used for meteorological variables other than solar radiation
- ▶ **year\_min:**
  - **description:** First year of the calculations
- ▶ **year\_max:**
  - **description:** Last year of the calculations
- ▶ **use\_horizon:**
  - **description:** Include horizon shadows
- ▶ **horizon\_db:**
  - **description:** Source of horizon data
- **mounting\_system:**
  - **description:** Mounting system
  - **choices:** fixed, vertical\_axis, inclined\_axis, two\_axis
  - **fields:**
    - ▶ **slope:**
      - **description:** Inclination angle from the horizontal plane
      - **units:** degree
    - ▶ **azimuth:**
      - **description:** Orientation (azimuth) angle of the (fixed) PV system (0 = S, 90 = W, -90 = E)
      - **units:** degree
- **pv\_module:**
  - **description:** PV module parameters
  - **variables:**
    - ▶ **technology:**
      - **description:** PV technology
    - ▶ **peak\_power:**
      - **description:** Nominal (peak) power of the PV module
      - **units:** kW
    - ▶ **system\_loss:**
      - **description:** Sum of system losses
      - **units:** %
- **economic\_data:**
  - **description:** Economic inputs
  - **variables:**
    - ▶ **system\_cost:**
      - **description:** Total cost of the PV system
      - **units:** user-defined currency
    - ▶ **interest:**
      - **description:** Annual interest
      - **units:** %/y
    - ▶ **lifetime:**
      - **description:** Expected lifetime of the PV system
      - **units:** y

- ▶ **outputs:**
  - **monthly:**
    - **type:** time series
    - **timestamp:** monthly averages
    - **variables:**
      - ▶ **E\_d:**
        - **description:** Average daily energy production from the given system
        - **units:** kWh/d
      - ▶ **E\_m:**
        - **description:** Average monthly energy production from the given system
        - **units:** kWh/mo
      - ▶ **H(i)\_d:**
        - **description:** Average daily sum of global irradiation per square meter received by the modules of the given system
        - **units:** kWh/m2/d
      - ▶ **H(i)\_m:**
        - **description:** Average monthly sum of global irradiation per square meter received by the modules of the given system
        - **units:** kWh/m2/mo
      - ▶ **SD\_m:**
        - **description:** Standard deviation of the monthly energy production due to year-to-year variation
        - **units:** kWh
  - **totals:**
    - **type:** time series totals
    - **variables:**
      - ▶ **E\_d:**
        - **description:** Average daily energy production from the given system
        - **units:** kWh/d
      - ▶ **E\_m:**
        - **description:** Average monthly energy production from the given system
        - **units:** kWh/mo
      - ▶ **E\_y:**
        - **description:** Average annual energy production from the given system
        - **units:** kWh/y
      - ▶ **H(i)\_d:**
        - **description:** Average daily sum of global irradiation per square meter received by the modules of the given system
        - **units:** kWh/m2/d
      - ▶ **H(i)\_m:**
        - **description:** Average monthly sum of global irradiation per square meter received by the modules of the given system

- **units:** kWh/m2/mo
- ▶ **H(i)\_y:**
  - **description:** Average annual sum of global irradiation per square meter received by the modules of the given system
  - **units:** kWh/m2/y
- ▶ **SD\_m:**
  - **description:** Standard deviation of the monthly energy production due to year-to-year variation
  - **units:** kWh
- ▶ **SD\_y:**
  - **description:** Standard deviation of the annual energy production due to year-to-year variation
  - **units:** kWh
- ▶ **I\_aoi:**
  - **description:** Angle of incidence loss
  - **units:** %
- ▶ **I\_spec:**
  - **description:** Spectral loss
  - **units:** %
- ▶ **I\_tg:**
  - **description:** Temperature and irradiance loss
  - **units:** %
- ▶ **I\_total:**
  - **description:** Total loss
  - **units:** %
- ▶ **LCOE\_pv:**
  - **description:** Levelized cost of the PV electricity
  - **units:** system\_cost\_currency/kWh

9.2. G4

- **inputs:**
  - **location:**
    - **latitude:** 36.666579105549324
    - **longitude:** −4.4610378208142265
    - **elevation:** 3.0
  - **meteo\_data:**
    - **radiation\_db:** PVGIS-SARAH3
    - **meteo\_db:** ERA5
    - **year\_min:** 2005
    - **year\_max:** 2023
    - **use\_horizon:** True
    - **horizon\_db:** DEM-calculated
  - **mounting\_system:**
    - **fixed:**
      - **slope:**
        - **value:** 0
        - **optimal:** False
      - **azimuth:**
        - **value:** 0
        - **optimal:** False
      - **type:** free-standing
  - **pv\_module:**
    - **technology:** c-Si
    - **peak\_power:** 2.7
    - **system\_loss:** 14.0
  - **economic\_data:**
    - **system\_cost:** 2700.0
    - **interest:** 5.0
    - **lifetime:** 20
- **outputs:**
  - **monthly:**
    - **fixed:**
      - **month:** 1
      - **E\_d:** 5.84
      - **E\_m:** 181.03
      - **H(i)\_d:** 2.78
      - **H(i)\_m:** 86.25
      - **SD\_m:** 16.25
      - **month:** 2
      - **E\_d:** 7.78
      - **E\_m:** 217.98
      - **H(i)\_d:** 3.63
      - **H(i)\_m:** 101.69
      - **SD\_m:** 20.6
- **month:** 3
  - **E\_d:** 10.01
  - **E\_m:** 310.45
  - **H(i)\_d:** 4.68
  - **H(i)\_m:** 145.12
  - **SD\_m:** 35.76
- **month:** 4
  - **E\_d:** 12.43
  - **E\_m:** 372.8
  - **H(i)\_d:** 5.87
  - **H(i)\_m:** 176.25
  - **SD\_m:** 28.3
- **month:** 5
  - **E\_d:** 14.69
  - **E\_m:** 455.28
  - **H(i)\_d:** 7.06
  - **H(i)\_m:** 218.93
  - **SD\_m:** 30.74
- **month:** 6
  - **E\_d:** 16.03
  - **E\_m:** 480.98
  - **H(i)\_d:** 7.85
  - **H(i)\_m:** 235.42
  - **SD\_m:** 15.75
- **month:** 7
  - **E\_d:** 15.77
  - **E\_m:** 488.76
  - **H(i)\_d:** 7.84
  - **H(i)\_m:** 243.17
  - **SD\_m:** 10.1
- **month:** 8
  - **E\_d:** 14.04
  - **E\_m:** 435.38
  - **H(i)\_d:** 6.97
  - **H(i)\_m:** 216.12
  - **SD\_m:** 12.81
- **month:** 9
  - **E\_d:** 11.35
  - **E\_m:** 340.45
  - **H(i)\_d:** 5.54
  - **H(i)\_m:** 166.16
  - **SD\_m:** 14.88
- **month:** 10
  - **E\_d:** 8.39
  - **E\_m:** 259.99
  - **H(i)\_d:** 4.06
  - **H(i)\_m:** 125.83
- **SD\_m:** 18.92
- **month:** 11
  - **E\_d:** 6.18
  - **E\_m:** 185.39
  - **H(i)\_d:** 2.96
  - **H(i)\_m:** 88.76
  - **SD\_m:** 15.98
- **month:** 12
  - **E\_d:** 5.16
  - **E\_m:** 159.81
  - **H(i)\_d:** 2.48
  - **H(i)\_m:** 76.97
  - **SD\_m:** 12.9
- **totals:**
  - **fixed:**
    - **E\_d:** 10.65
    - **E\_m:** 324.03
    - **E\_y:** 3888.3
    - **H(i)\_d:** 5.15
    - **H(i)\_m:** 156.72
    - **H(i)\_y:** 1880.65
    - **SD\_m:** 6.34
    - **SD\_y:** 76.05
    - **l\_aoi:** −3.51
    - **l\_spec:** ?(0)
    - **l\_tg:** −7.72
    - **l\_total:** −23.42
    - **LCOE\_pv:** 0.07
- **meta:**
  - **inputs:**
    - **location:**
      - **description:** Selected location
    - **variables:**
      - **latitude:**
        - **description:** Latitude
        - **units:** decimal degree
      - **longitude:**
        - **description:** Longitude
        - **units:** decimal degree
      - **elevation:**
        - **description:** Elevation
        - **units:** m
    - **meteo\_data:**
      - **description:** Sources of meteorological data
      - **variables:**
        - **radiation\_db:**
          - **description:** Solar radiation database

- **meteo\_db:**
  - **description:** Database used for meteorological variables other than solar radiation
- **year\_min:**
  - **description:** First year of the calculations
- **year\_max:**
  - **description:** Last year of the calculations
- **use\_horizon:**
  - **description:** Include horizon shadows
- **horizon\_db:**
  - **description:** Source of horizon data
- **mounting\_system:**
  - **description:** Mounting system
  - **choices:** fixed, vertical\_axis, inclined\_axis, two\_axis
  - **fields:**
    - **slope:**
      - **description:** Inclination angle from the horizontal plane
      - **units:** degree
    - **azimuth:**
      - **description:** Orientation (azimuth) angle of the (fixed) PV system (0 = S, 90 = W, -90 = E)
      - **units:** degree
- **pv\_module:**
  - **description:** PV module parameters
  - **variables:**
    - **technology:**
      - **description:** PV technology
    - **peak\_power:**
      - **description:** Nominal (peak) power of the PV module
      - **units:** kW
    - **system\_loss:**
      - **description:** Sum of system losses
      - **units:** %
- **economic\_data:**
  - **description:** Economic inputs
  - **variables:**
    - **system\_cost:**
      - **description:** Total cost of the PV system
      - **units:** user-defined currency
    - **interest:**
      - **description:** Annual interest
      - **units:** %/y
    - **lifetime:**
      - **description:** Expected lifetime of the PV system
      - **units:** y
- **outputs:**
  - **monthly:**

- **type:** time series
- **timestamp:** monthly averages
- **variables:**
  - **E\_d:**
    - **description:** Average daily energy production from the given system
    - **units:** kWh/d
  - **E\_m:**
    - **description:** Average monthly energy production from the given system
    - **units:** kWh/mo
  - **H(i)\_d:**
    - **description:** Average daily sum of global irradiation per square meter received by the modules of the given system
    - **units:** kWh/m2/d
  - **H(i)\_m:**
    - **description:** Average monthly sum of global irradiation per square meter received by the modules of the given system
    - **units:** kWh/m2/mo
  - **SD\_m:**
    - **description:** Standard deviation of the monthly energy production due to year-to-year variation
    - **units:** kWh
- **totals:**
  - **type:** time series totals
  - **variables:**
    - **E\_d:**
      - **description:** Average daily energy production from the given system
      - **units:** kWh/d
    - **E\_m:**
      - **description:** Average monthly energy production from the given system
      - **units:** kWh/mo
    - **E\_y:**
      - **description:** Average annual energy production from the given system
      - **units:** kWh/y
    - **H(i)\_d:**
      - **description:** Average daily sum of global irradiation per square meter received by the modules of the given system
      - **units:** kWh/m2/d
    - **H(i)\_m:**
      - **description:** Average monthly sum of global irradiation per square meter received by the modules of the given system
      - **units:** kWh/m2/mo
    - **H(i)\_y:**

- **description:** Average annual sum of global irradiation per square meter received by the modules of the given system
- **units:** kWh/m2/y
- **SD\_m:**
  - **description:** Standard deviation of the monthly energy production due to year-to-year variation
  - **units:** kWh
- **SD\_y:**
  - **description:** Standard deviation of the annual energy production due to year-to-year variation
  - **units:** kWh
- **l\_aoi:**
  - **description:** Angle of incidence loss
  - **units:** %
- **l\_spec:**
  - **description:** Spectral loss
  - **units:** %
- **l\_tg:**
  - **description:** Temperature and irradiance loss
  - **units:** %
- **l\_total:**
  - **description:** Total loss
  - **units:** %
- **LCOE\_pv:**
  - **description:** Levelized cost of the PV electricity
  - **units:** system\_cost\_currency/kWh

9.3. G3

- **inputs:**
  - **location:**
    - **latitude:** 36.666579105549324
    - **longitude:** −4.4610378208142265
    - **elevation:** 3.0
  - **meteo\_data:**
    - **radiation\_db:** PVGIS-SARAH3
    - **meteo\_db:** ERA5
    - **year\_min:** 2005
    - **year\_max:** 2023
    - **use\_horizon:** True
    - **horizon\_db:** DEM-calculated
  - **mounting\_system:**
    - **fixed:**
      - **slope:**
        - **value:** 26
        - **optimal:** False
      - **azimuth:**
        - **value:** 0
        - **optimal:** False
      - **type:** free-standing
  - **pv\_module:**
    - **technology:** c-Si
    - **peak\_power:** 2.0
    - **system\_loss:** 14.0
  - **economic\_data:**
    - **system\_cost:** 1998.0
    - **interest:** 5.0
    - **lifetime:** 20
- **outputs:**
  - **monthly:**
    - **fixed:**
      - **month:** 1
      - **E\_d:** 7.05
      - **E\_m:** 218.63
      - **H(i)\_d:** 4.36
      - **H(i)\_m:** 135.12
      - **SD\_m:** 23.66
      - **month:** 2
      - **E\_d:** 7.99
      - **E\_m:** 223.73
      - **H(i)\_d:** 4.96
      - **H(i)\_m:** 138.89
      - **SD\_m:** 25.75

- **month:** 3
- **E\_d:** 8.83
- **E\_m:** 273.62
- **H(i)\_d:** 5.58
- **H(i)\_m:** 172.89
- **SD\_m:** 36.06
- **month:** 4
- **E\_d:** 9.75
- **E\_m:** 292.57
- **H(i)\_d:** 6.26
- **H(i)\_m:** 187.77
- **SD\_m:** 24.03
- **month:** 5
- **E\_d:** 10.62
- **E\_m:** 329.25
- **H(i)\_d:** 6.94
- **H(i)\_m:** 215.12
- **SD\_m:** 22.39
- **month:** 6
- **E\_d:** 11.16
- **E\_m:** 334.83
- **H(i)\_d:** 7.42
- **H(i)\_m:** 222.69
- **SD\_m:** 10.9
- **month:** 7
- **E\_d:** 11.19
- **E\_m:** 346.86
- **H(i)\_d:** 7.57
- **H(i)\_m:** 234.63
- **SD\_m:** 7.06
- **month:** 8
- **E\_d:** 10.72
- **E\_m:** 332.17
- **H(i)\_d:** 7.23
- **H(i)\_m:** 224.23
- **SD\_m:** 10.08
- **month:** 9
- **E\_d:** 9.63
- **E\_m:** 288.78
- **H(i)\_d:** 6.37
- **H(i)\_m:** 191.21
- **SD\_m:** 14.6
- **month:** 10
- **E\_d:** 8.09
- **E\_m:** 250.81
- **H(i)\_d:** 5.26
- **H(i)\_m:** 162.99

- **SD\_m:** 22.29
- **month:** 11
- **E\_d:** 6.95
- **E\_m:** 208.61
- **H(i)\_d:** 4.37
- **H(i)\_m:** 130.97
- **SD\_m:** 21.7
- **month:** 12
- **E\_d:** 6.46
- **E\_m:** 200.14
- **H(i)\_d:** 3.98
- **H(i)\_m:** 123.46
- **SD\_m:** 20.13
- **totals:**
  - **fixed:**
    - **E\_d:** 9.04
    - **E\_m:** 275.0
    - **E\_y:** 3300.01
    - **H(i)\_d:** 5.86
    - **H(i)\_m:** 178.33
    - **H(i)\_y:** 2139.95
    - **SD\_m:** 6.91
    - **SD\_y:** 82.94
    - **l\_aoi:** −2.65
    - **l\_spec:** ?(0)
    - **l\_tg:** −7.9
    - **l\_total:** −22.9
    - **LCOE\_pv:** 0.06
- **meta:**
  - **inputs:**
    - **location:**
      - **description:** Selected location
    - **variables:**
      - **latitude:**
        - **description:** Latitude
        - **units:** decimal degree
      - **longitude:**
        - **description:** Longitude
        - **units:** decimal degree
      - **elevation:**
        - **description:** Elevation
        - **units:** m
    - **meteo\_data:**
      - **description:** Sources of meteorological data
      - **variables:**
        - **radiation\_db:**
          - **description:** Solar radiation database

- **meteo\_db:**
  - **description:** Database used for meteorological variables other than solar radiation
- **year\_min:**
  - **description:** First year of the calculations
- **year\_max:**
  - **description:** Last year of the calculations
- **use\_horizon:**
  - **description:** Include horizon shadows
- **horizon\_db:**
  - **description:** Source of horizon data
- **mounting\_system:**
  - **description:** Mounting system
  - **choices:** fixed, vertical\_axis, inclined\_axis, two\_axis
  - **fields:**
    - **slope:**
      - **description:** Inclination angle from the horizontal plane
      - **units:** degree
    - **azimuth:**
      - **description:** Orientation (azimuth) angle of the (fixed) PV system (0 = S, 90 = W, -90 = E)
      - **units:** degree
- **pv\_module:**
  - **description:** PV module parameters
  - **variables:**
    - **technology:**
      - **description:** PV technology
    - **peak\_power:**
      - **description:** Nominal (peak) power of the PV module
      - **units:** kW
    - **system\_loss:**
      - **description:** Sum of system losses
      - **units:** %
- **economic\_data:**
  - **description:** Economic inputs
  - **variables:**
    - **system\_cost:**
      - **description:** Total cost of the PV system
      - **units:** user-defined currency
    - **interest:**
      - **description:** Annual interest
      - **units:** %/y
    - **lifetime:**
      - **description:** Expected lifetime of the PV system
      - **units:** y
- **outputs:**
  - **monthly:**

- **type:** time series
- **timestamp:** monthly averages
- **variables:**
  - **E\_d:**
    - **description:** Average daily energy production from the given system
    - **units:** kWh/d
  - **E\_m:**
    - **description:** Average monthly energy production from the given system
    - **units:** kWh/mo
  - **H(i)\_d:**
    - **description:** Average daily sum of global irradiation per square meter received by the modules of the given system
    - **units:** kWh/m2/d
  - **H(i)\_m:**
    - **description:** Average monthly sum of global irradiation per square meter received by the modules of the given system
    - **units:** kWh/m2/mo
  - **SD\_m:**
    - **description:** Standard deviation of the monthly energy production due to year-to-year variation
    - **units:** kWh
- **totals:**
  - **type:** time series totals
  - **variables:**
    - **E\_d:**
      - **description:** Average daily energy production from the given system
      - **units:** kWh/d
    - **E\_m:**
      - **description:** Average monthly energy production from the given system
      - **units:** kWh/mo
    - **E\_y:**
      - **description:** Average annual energy production from the given system
      - **units:** kWh/y
    - **H(i)\_d:**
      - **description:** Average daily sum of global irradiation per square meter received by the modules of the given system
      - **units:** kWh/m2/d
    - **H(i)\_m:**
      - **description:** Average monthly sum of global irradiation per square meter received by the modules of the given system
      - **units:** kWh/m2/mo
    - **H(i)\_y:**

- **description:** Average annual sum of global irradiation per square meter received by the modules of the given system
- **units:** kWh/m2/y
- **SD\_m:**
  - **description:** Standard deviation of the monthly energy production due to year-to-year variation
  - **units:** kWh
- **SD\_y:**
  - **description:** Standard deviation of the annual energy production due to year-to-year variation
  - **units:** kWh
- **l\_aoi:**
  - **description:** Angle of incidence loss
  - **units:** %
- **l\_spec:**
  - **description:** Spectral loss
  - **units:** %
- **l\_tg:**
  - **description:** Temperature and irradiance loss
  - **units:** %
- **l\_total:**
  - **description:** Total loss
  - **units:** %
- **LCOE\_pv:**
  - **description:** Levelized cost of the PV electricity
  - **units:** system\_cost\_currency/kWh

9.4. G2

- **inputs:**
  - **location:**
    - **latitude:** 36.666579105549324
    - **longitude:** −4.4610378208142265
    - **elevation:** 3.0
  - **meteo\_data:**
    - **radiation\_db:** PVGIS-SARAH3
    - **meteo\_db:** ERA5
    - **year\_min:** 2005
    - **year\_max:** 2023
    - **use\_horizon:** True
    - **horizon\_db:** DEM-calculated
  - **mounting\_system:**
    - **fixed:**
      - **slope:**
        - **value:** 0
        - **optimal:** False
      - **azimuth:**
        - **value:** 0
        - **optimal:** False
      - **type:** free-standing
  - **pv\_module:**
    - **technology:** c-Si
    - **peak\_power:** 3.24
    - **system\_loss:** 14.0
  - **economic\_data:**
    - **system\_cost:** 3240.0
    - **interest:** 5.0
    - **lifetime:** 20
- **outputs:**
  - **monthly:**
    - **fixed:**
      - **month:** 1
      - **E\_d:** 7.01
      - **E\_m:** 217.24
      - **H(i)\_d:** 2.78
      - **H(i)\_m:** 86.25
      - **SD\_m:** 19.5
      - **month:** 2
      - **E\_d:** 9.34
      - **E\_m:** 261.57
      - **H(i)\_d:** 3.63
      - **H(i)\_m:** 101.69
      - **SD\_m:** 24.72

- **month:** 3
- **E\_d:** 12.02
- **E\_m:** 372.54
- **H(i)\_d:** 4.68
- **H(i)\_m:** 145.12
- **SD\_m:** 42.92
- **month:** 4
- **E\_d:** 14.91
- **E\_m:** 447.36
- **H(i)\_d:** 5.87
- **H(i)\_m:** 176.25
- **SD\_m:** 33.96
- **month:** 5
- **E\_d:** 17.62
- **E\_m:** 546.33
- **H(i)\_d:** 7.06
- **H(i)\_m:** 218.93
- **SD\_m:** 36.88
- **month:** 6
- **E\_d:** 19.24
- **E\_m:** 577.18
- **H(i)\_d:** 7.85
- **H(i)\_m:** 235.42
- **SD\_m:** 18.89
- **month:** 7
- **E\_d:** 18.92
- **E\_m:** 586.51
- **H(i)\_d:** 7.84
- **H(i)\_m:** 243.17
- **SD\_m:** 12.13
- **month:** 8
- **E\_d:** 16.85
- **E\_m:** 522.46
- **H(i)\_d:** 6.97
- **H(i)\_m:** 216.12
- **SD\_m:** 15.38
- **month:** 9
- **E\_d:** 13.62
- **E\_m:** 408.54
- **H(i)\_d:** 5.54
- **H(i)\_m:** 166.16
- **SD\_m:** 17.86
- **month:** 10
- **E\_d:** 10.06
- **E\_m:** 311.99
- **H(i)\_d:** 4.06
- **H(i)\_m:** 125.83

- **SD\_m:** 22.71
- **month:** 11
- **E\_d:** 7.42
- **E\_m:** 222.47
- **H(i)\_d:** 2.96
- **H(i)\_m:** 88.76
- **SD\_m:** 19.18
- **month:** 12
- **E\_d:** 6.19
- **E\_m:** 191.77
- **H(i)\_d:** 2.48
- **H(i)\_m:** 76.97
- **SD\_m:** 15.48
- **totals:**
  - **fixed:**
    - **E\_d:** 12.78
    - **E\_m:** 388.83
    - **E\_y:** 4665.96
    - **H(i)\_d:** 5.15
    - **H(i)\_m:** 156.72
    - **H(i)\_y:** 1880.65
    - **SD\_m:** 7.61
    - **SD\_y:** 91.26
    - **l\_aoi:** −3.51
    - **l\_spec:** ?(0)
    - **l\_tg:** −7.72
    - **l\_total:** −23.42
    - **LCOE\_pv:** 0.07
- **meta:**
  - **inputs:**
    - **location:**
      - **description:** Selected location
    - **variables:**
      - **latitude:**
        - **description:** Latitude
        - **units:** decimal degree
      - **longitude:**
        - **description:** Longitude
        - **units:** decimal degree
      - **elevation:**
        - **description:** Elevation
        - **units:** m
    - **meteo\_data:**
      - **description:** Sources of meteorological data
      - **variables:**
        - **radiation\_db:**
          - **description:** Solar radiation database

- **meteo\_db:**
  - **description:** Database used for meteorological variables other than solar radiation
- **year\_min:**
  - **description:** First year of the calculations
- **year\_max:**
  - **description:** Last year of the calculations
- **use\_horizon:**
  - **description:** Include horizon shadows
- **horizon\_db:**
  - **description:** Source of horizon data
- **mounting\_system:**
  - **description:** Mounting system
  - **choices:** fixed, vertical\_axis, inclined\_axis, two\_axis
  - **fields:**
    - **slope:**
      - **description:** Inclination angle from the horizontal plane
      - **units:** degree
    - **azimuth:**
      - **description:** Orientation (azimuth) angle of the (fixed) PV system (0 = S, 90 = W, -90 = E)
      - **units:** degree
- **pv\_module:**
  - **description:** PV module parameters
  - **variables:**
    - **technology:**
      - **description:** PV technology
    - **peak\_power:**
      - **description:** Nominal (peak) power of the PV module
      - **units:** kW
    - **system\_loss:**
      - **description:** Sum of system losses
      - **units:** %
- **economic\_data:**
  - **description:** Economic inputs
  - **variables:**
    - **system\_cost:**
      - **description:** Total cost of the PV system
      - **units:** user-defined currency
    - **interest:**
      - **description:** Annual interest
      - **units:** %/y
    - **lifetime:**
      - **description:** Expected lifetime of the PV system
      - **units:** y
- **outputs:**
  - **monthly:**

- **type:** time series
- **timestamp:** monthly averages
- **variables:**
  - **E\_d:**
    - **description:** Average daily energy production from the given system
    - **units:** kWh/d
  - **E\_m:**
    - **description:** Average monthly energy production from the given system
    - **units:** kWh/mo
  - **H(i)\_d:**
    - **description:** Average daily sum of global irradiation per square meter received by the modules of the given system
    - **units:** kWh/m2/d
  - **H(i)\_m:**
    - **description:** Average monthly sum of global irradiation per square meter received by the modules of the given system
    - **units:** kWh/m2/mo
  - **SD\_m:**
    - **description:** Standard deviation of the monthly energy production due to year-to-year variation
    - **units:** kWh
- **totals:**
  - **type:** time series totals
  - **variables:**
    - **E\_d:**
      - **description:** Average daily energy production from the given system
      - **units:** kWh/d
    - **E\_m:**
      - **description:** Average monthly energy production from the given system
      - **units:** kWh/mo
    - **E\_y:**
      - **description:** Average annual energy production from the given system
      - **units:** kWh/y
    - **H(i)\_d:**
      - **description:** Average daily sum of global irradiation per square meter received by the modules of the given system
      - **units:** kWh/m2/d
    - **H(i)\_m:**
      - **description:** Average monthly sum of global irradiation per square meter received by the modules of the given system
      - **units:** kWh/m2/mo
    - **H(i)\_y:**

- **description:** Average annual sum of global irradiation per square meter received by the modules of the given system
- **units:** kWh/m2/y
- **SD\_m:**
  - **description:** Standard deviation of the monthly energy production due to year-to-year variation
  - **units:** kWh
- **SD\_y:**
  - **description:** Standard deviation of the annual energy production due to year-to-year variation
  - **units:** kWh
- **l\_aoi:**
  - **description:** Angle of incidence loss
  - **units:** %
- **l\_spec:**
  - **description:** Spectral loss
  - **units:** %
- **l\_tg:**
  - **description:** Temperature and irradiance loss
  - **units:** %
- **l\_total:**
  - **description:** Total loss
  - **units:** %
- **LCOE\_pv:**
  - **description:** Levelized cost of the PV electricity
  - **units:** system\_cost\_currency/kWh

9.5. Total

- **inputs:**
  - **location:**
    - **latitude:** 36.67
    - **longitude:** −4.46
    - **elevation:** 3.0
  - **meteo\_data:**
    - **radiation\_db:** PVGIS-SARAH3
    - **meteo\_db:** ERA5
    - **year\_min:** 2005
    - **year\_max:** 2023
    - **use\_horizon:** True
    - **horizon\_db:** DEM-calculated
  - **mounting\_system:**
    - **fixed:**
      - **slope:**
        - **value:** 0
        - **optimal:** False
      - **azimuth:**
        - **value:** 0
        - **optimal:** False
      - **type:** free-standing
  - **pv\_module:**
    - **technology:** c-Si
    - **peak\_power:** 14.42
    - **system\_loss:** 14.0
  - **economic\_data:**
    - **system\_cost:** 3240.0
    - **interest:** 5.0
    - **lifetime:** 20
- **outputs:**
  - **monthly:**
    - **fixed:**
      - **month:** 1
      - **E\_d:** 33.92
      - **E\_m:** 1051.37
      - **H(i)\_d:** 3.0
      - **H(i)\_m:** 93.03
      - **SD\_m:** 98.41
      - **month:** 2
      - **E\_d:** 43.79
      - **E\_m:** 1226.42
      - **H(i)\_d:** 3.81
      - **H(i)\_m:** 106.85
      - **SD\_m:** 120.5

- **month:** 3
- **E\_d:** 54.9
- **E\_m:** 1701.7
- **H(i)\_d:** 4.8
- **H(i)\_m:** 148.97
- **SD\_m:** 200.58
- **month:** 4
- **E\_d:** 66.91
- **E\_m:** 2007.45
- **H(i)\_d:** 5.92
- **H(i)\_m:** 177.85
- **SD\_m:** 154.22
- **month:** 5
- **E\_d:** 78.18
- **E\_m:** 2423.52
- **H(i)\_d:** 7.04
- **H(i)\_m:** 218.4
- **SD\_m:** 163.78
- **month:** 6
- **E\_d:** 84.91
- **E\_m:** 2547.34
- **H(i)\_d:** 7.79
- **H(i)\_m:** 233.65
- **SD\_m:** 83.33
- **month:** 7
- **E\_d:** 83.72
- **E\_m:** 2595.16
- **H(i)\_d:** 7.8
- **H(i)\_m:** 241.99
- **SD\_m:** 53.54
- **month:** 8
- **E\_d:** 75.32
- **E\_m:** 2334.93
- **H(i)\_d:** 7.01
- **H(i)\_m:** 217.24
- **SD\_m:** 69.02
- **month:** 9
- **E\_d:** 61.84
- **E\_m:** 1854.85
- **H(i)\_d:** 5.66
- **H(i)\_m:** 169.63
- **SD\_m:** 83.06
- **month:** 10
- **E\_d:** 46.67
- **E\_m:** 1446.78
- **H(i)\_d:** 4.23
- **H(i)\_m:** 130.98

- **SD\_m:** 109.33
- **month:** 11
- **E\_d:** 35.38
- **E\_m:** 1061.42
- **H(i)\_d:** 3.16
- **H(i)\_m:** 94.61
- **SD\_m:** 95.22
- **month:** 12
- **E\_d:** 30.18
- **E\_m:** 935.25
- **H(i)\_d:** 2.69
- **H(i)\_m:** 83.42
- **SD\_m:** 79.46
- **totals:**
  - **fixed:**
    - **E\_d:** 58.04
    - **E\_m:** 1765.52
    - **E\_y:** 21186.19
    - **H(i)\_d:** 5.25
    - **H(i)\_m:** 159.72
    - **H(i)\_y:** 1916.61
    - **SD\_m:** 36.07
    - **SD\_y:** 432.77
    - **l\_aoi:** −3.39
    - **l\_spec:** 0.0
    - **l\_tg:** −7.74
    - **l\_total:** −23.35
    - **LCOE\_pv:** 0.07
- **meta:**
  - **inputs:**
    - **location:**
      - **description:** Selected location
    - **variables:**
      - **latitude:**
        - **description:** Latitude
        - **units:** decimal degree
      - **longitude:**
        - **description:** Longitude
        - **units:** decimal degree
      - **elevation:**
        - **description:** Elevation
        - **units:** m
    - **meteo\_data:**
      - **description:** Sources of meteorological data
      - **variables:**
        - **radiation\_db:**
          - **description:** Solar radiation database

- **meteo\_db:**
  - **description:** Database used for meteorological variables other than solar radiation
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- **year\_max:**
  - **description:** Last year of the calculations
- **use\_horizon:**
  - **description:** Include horizon shadows
- **horizon\_db:**
  - **description:** Source of horizon data
- **mounting\_system:**
  - **description:** Mounting system
  - **choices:** fixed, vertical\_axis, inclined\_axis, two\_axis
  - **fields:**
    - **slope:**
      - **description:** Inclination angle from the horizontal plane
      - **units:** degree
    - **azimuth:**
      - **description:** Orientation (azimuth) angle of the (fixed) PV system (0 = S, 90 = W, -90 = E)
      - **units:** degree
- **pv\_module:**
  - **description:** PV module parameters
  - **variables:**
    - **technology:**
      - **description:** PV technology
    - **peak\_power:**
      - **description:** Nominal (peak) power of the PV module
      - **units:** kW
    - **system\_loss:**
      - **description:** Sum of system losses
      - **units:** %
- **economic\_data:**
  - **description:** Economic inputs
  - **variables:**
    - **system\_cost:**
      - **description:** Total cost of the PV system
      - **units:** user-defined currency
    - **interest:**
      - **description:** Annual interest
      - **units:** %/y
    - **lifetime:**
      - **description:** Expected lifetime of the PV system
      - **units:** y
- **outputs:**
  - **monthly:**

- **type:** time series
- **timestamp:** monthly averages
- **variables:**
  - **E\_d:**
    - **description:** Average daily energy production from the given system
    - **units:** kWh/d
  - **E\_m:**
    - **description:** Average monthly energy production from the given system
    - **units:** kWh/mo
  - **H(i)\_d:**
    - **description:** Average daily sum of global irradiation per square meter received by the modules of the given system
    - **units:** kWh/m2/d
  - **H(i)\_m:**
    - **description:** Average monthly sum of global irradiation per square meter received by the modules of the given system
    - **units:** kWh/m2/mo
  - **SD\_m:**
    - **description:** Standard deviation of the monthly energy production due to year-to-year variation
    - **units:** kWh
- **totals:**
  - **type:** time series totals
  - **variables:**
    - **E\_d:**
      - **description:** Average daily energy production from the given system
      - **units:** kWh/d
    - **E\_m:**
      - **description:** Average monthly energy production from the given system
      - **units:** kWh/mo
    - **E\_y:**
      - **description:** Average annual energy production from the given system
      - **units:** kWh/y
    - **H(i)\_d:**
      - **description:** Average daily sum of global irradiation per square meter received by the modules of the given system
      - **units:** kWh/m2/d
    - **H(i)\_m:**
      - **description:** Average monthly sum of global irradiation per square meter received by the modules of the given system
      - **units:** kWh/m2/mo
    - **H(i)\_y:**

- **description:** Average annual sum of global irradiation per square meter received by the modules of the given system
- **units:** kWh/m2/y
- **SD\_m:**
  - **description:** Standard deviation of the monthly energy production due to year-to-year variation
  - **units:** kWh
- **SD\_y:**
  - **description:** Standard deviation of the annual energy production due to year-to-year variation
  - **units:** kWh
- **l\_aoi:**
  - **description:** Angle of incidence loss
  - **units:** %
- **l\_spec:**
  - **description:** Spectral loss
  - **units:** %
- **l\_tg:**
  - **description:** Temperature and irradiance loss
  - **units:** %
- **l\_total:**
  - **description:** Total loss
  - **units:** %
- **LCOE\_pv:**
  - **description:** Levelized cost of the PV electricity
  - **units:** system\_cost\_currency/kWh

